

ezclean-data: API Reference & Guide

A comprehensive, dataset-agnostic Python library for automated data ingestion, cleaning, preprocessing, and exploratory data analysis.

Project Links:

- **PyPI Registry:** <https://pypi.org/project/ezclean-data/>
 - **Source Repository:** <https://github.com/Thilac01/ezclean>
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1 Quick Start

The following workflow demonstrates the automated data loading, pipeline execution, analytical visualization, and self-contained dashboard creation features using native abstractions:

```
from ezclean import Smart_loader, Cleaner, colname, plot, plot_dashboard

# Load dataset
df = Smart_loader("tested.csv")

# Execute cleaning pipeline
df_cleaned = Cleaner(df)

# Display column statistics
colname(df_cleaned)

# Visualize a single column
plot(df_cleaned, "survived")

# Generate a relationship matrix
plot(df_cleaned)

# Create an interactive dashboard
plot_dashboard(
    df_cleaned,
    filename="my_dashboard.html"
)
```

2 API Reference

`Smart_loader(file_path, **kwargs)`

Automatically loads structured datasets from local storage or remote URLs.

Supported Formats

Category	Formats
Text Files	CSV, TSV, TXT
JSON Formats	JSON, JSONL, NDJSON
Spreadsheet Files	XLSX, XLS, ODS
Columnar Formats	Parquet, Feather, Arrow, ORC
Statistical Formats	SPSS, SAS, Stata
Other Formats	XML, HTML, Pickle, HDF

`Cleaner(df, ...)`

Executes a complete data-cleaning pipeline across the entire target data context.

Included Operations

- **Column Name Standardization:** Converts all names to structured **snake_case**, filters special syntax characters, and cleans out duplicate inner separators.
- **Data Sanitization:** Detects and seamlessly drops common low-quality string raw text flags such as `?`, `NULL`, `null`, `nil`, `N/A`, or `NaN`, mapping them to precise system missing-value markers.
- **Text Normalization:** Strips out trailing or leading space padding matrices and standardizes overall text string alignments.
- **Automatic Type Detection:** Implicitly forces elements to numeric columns where valid mathematical mappings hold true, while securely isolating incompatible data arrays.
- **Missing Value Handling:**
 - Numerical columns → Median Imputation
 - Categorical columns → "Unknown" Replacement
- **Outlier Treatment:** Finds and removes high-variance anomalous values using Interquartile Range (IQR) calculation bounds automatically.

`colname(df)`

Generates and displays comprehensive matrix summaries for all structural columns, providing rapid insight into metadata properties: 2

- Data type verification
- Absolute missing value counts
- Completeness percentage ratios
- Unique value counts
- Statistical summaries

`plot(df, target_column=None, columns=None)`

Single-Column Visualization

When a target identifier parameter is provided, the function picks the appropriate visual display mapping based on the incoming column structure:

Data Type	Visualization Engine Selected
Numeric	Histogram + Box Plot
Categorical	Bar Chart + Donut Chart
Datetime	Sequential Trend Line

Relationship Matrix Grid

When called as `plot(df)` without parameters, it automatically builds a comprehensive, high-dimensional pairplot graph matrix showing univariate tracking trends, localized data densities, correlations, and feature trends.



Figure 1: ezclean-data Complete Analytical Pairplot Matrix Displaying Cross-Variable Trend Grids

```
plot_dashboard(df, filename="ezclean_dashboard.html", show=True)
```

Compiles and generates a self-contained, interactive HTML exploratory dashboard file that can be shared and works entirely offline.

Dashboard Components

- **Dataset Summary Profile:** Tracks high-level row-column configurations, data point distribution matrices, and global quality health percentages.
- **Column Properties Analysis:** Lists isolated data structures, missing property trackers, and specific category metrics.
- **Interactive Plot Designer:** Allows developers and analysts to build custom plots dynamically via runtime dashboard controls by choosing specific X/Y axis options without changing any code.

3 Operational Example Workflow

Here is a comprehensive example script combining ingestion, cleaning, and reporting in a single processing step:

```
from ezclean import *

# Ingest heterogeneous dataset files cleanly
df = Smart_loader("data.csv")

# Clean standard string layouts, parse values, and handle outlier
  structures
df = Cleaner(df)
```

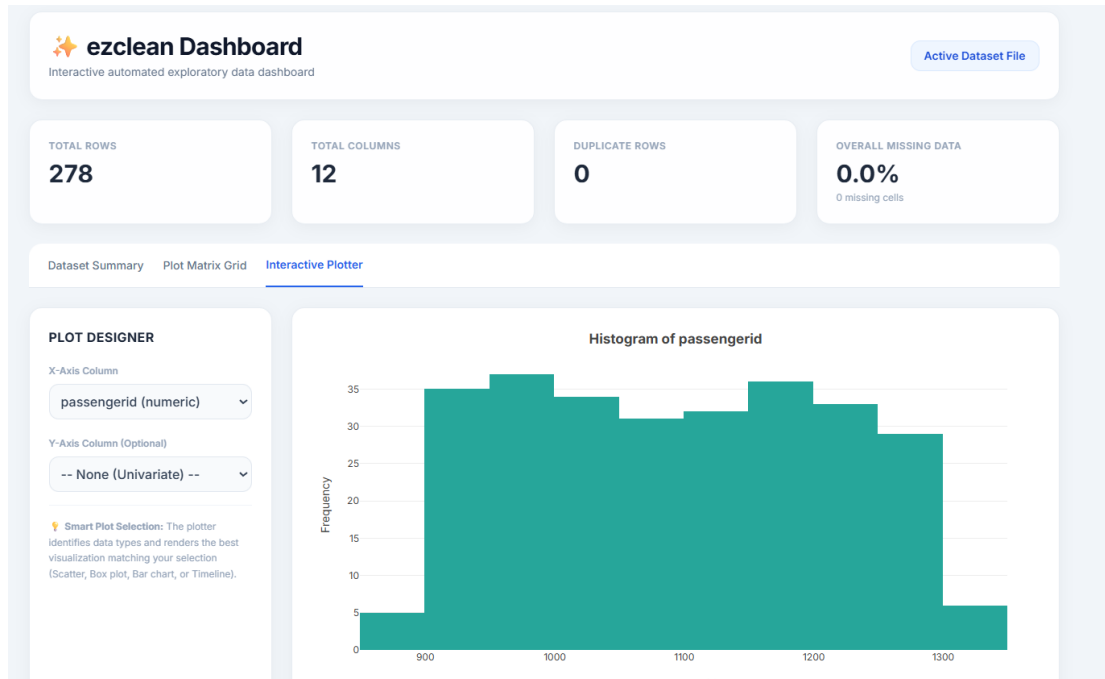


Figure 2: Interactive ezclean-data Exploration Dashboard UI Featuring Dynamic Plot Selection Tools

```
# Print structural validation updates
colname(df)

# Render numeric evaluations
plot(df, "age")

# Extract the cross-correlation summary grid layout
plot(df)

# Write output dashboard HTML artifact files
plot_dashboard(
  df,
  filename="dashboard.html"
)
```